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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech. (CBCS) Degree Examinations

December 2025

EE2001-1 – ELECTRICAL MACHINES - II

(For EEE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the 'OMR Sheet' provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The speed of a DC series motor is:**
 - Directly proportional to load applied
 - Inversely proportional to load applied
 - Independent of the load
 - Zero at no load
- Swinburne's test is conducted on**
 - Loaded DC motor
 - Loaded DC generator
 - Unloaded DC motor
 - Unloaded DC generator
- From the retardation test on the DC motor following is determined.**
 - Moment of inertia of armature
 - Speed of the motor
 - Armature resistance
 - Developed torque
- Hopkinson's test is also called:**
 - No load test
 - Sumpner's test
 - Retardation test
 - Heat run test
- The Limitation of Swinburne's test is:**
 - Cannot test under load
 - Two motors are required
 - DC series motor cannot be tested.
 - DC shunt motor cannot be tested.
- In the Field's test, field winding of the generator and motor are connected in series for the following reason.**
 - Ease of connectivity
 - To make the mechanical losses equal.
 - To reduce the copper losses
 - To make the iron losses of both the machines equal.
- Synchronous motor is also called as**
 - Synchronous condenser
 - Asynchronous motor
 - Work horse of the industry
 - Asynchronous condenser
- Main feature of Brush less DC motor is**
 - Electronic commutator without position sensitive controller
 - Mechanical commutator
 - Electronic commutator with position sensitive controller
 - Permanent magnets are used in rotor.
- Starters are used in DC motors to,**
 - To limit the high starting current
 - To increase the speed
 - To keep the speed constant
 - To provide the initial torque
- A 6-pole dc motor rotates by 12 degrees. what is the corresponding electrical angle covered?**
 - 4 degrees
 - 36 degrees
 - 72 degrees
 - 40 degrees

11. When the load is ZPF inductive, the armature reaction in a three-phase synchronous generator:
 - A) is demagnetising
 - B) is magnetising
 - C) is cross-magnetising
 - D) has no effect on induced emf.
12. Which of the following is called pessimistic method of regulation determination for a three-phase alternator?
 - A) MMF method
 - B) EMF method
 - C) ZPF method
 - D) None of the methods
13. Alternator on infinite bus bar has constant
 - A) Terminal voltage and frequency
 - B) Frequency
 - C) Power factor
 - D) Power factor and terminal voltage
14. Generated e.m.f. for same field current and double speed will be
 - A) same
 - B) 4 times
 - C) double
 - D) Half
15. Voltage regulation for an alternator operating at leading power factor is negative due to
 - A) magnetizing nature of armature reaction
 - B) demagnetizing nature of armature reaction
 - C) cross-magnetizing nature of armature reaction
 - D) all of the mentioned
16. The frequency of EMF generated by an alternator having 4 poles and rotating at 1800 RPM is
 - A) 450
 - B) 7200
 - C) 120
 - D) 60
17. Which of the following statements is true regarding EMF, MMF and ZPF methods of determining voltage regulation of alternator?
 - A) EMF method is the most accurate method
 - B) MMF method is called Potier triangle method
 - C) ZPF method is the most accurate
 - D) ZPF method is also called Synchronous impedance method
18. Some engineers prefer 'lamps bright' synchronization to 'lamps dark' synchronization because
 - A) brightness variation of lamps can be judged easily
 - B) Darkness of lamps can be judged easily
 - C) flicker is more pronounced
 - D) it can be performed slowly
19. Keeping the excitation constant, if steam supply of an alternator running in parallel with another identical alternator is increased, then
 - A) it would over-run the other alternator
 - B) it will supply greater portion of the load
 - C) its rotor will fall back in phase with respect to the other machine
 - D) its power factor would be decreased
20. For a salient pole alternator X_d/X_q ratio
 - A) Cannot be measured
 - B) Less than 1
 - C) Equal to 1
 - D) Greater than 1

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain different methods of speed control in DC series and dc shunt motors.	08	L*3	01	1,2,9,10,1
	b) Two shunt machines loaded for the Hopkinson's test take 15 A at 200 V from the supply. The motor current is 100 A and the shunt currents are 3 A and 2.5 A. If armature resistance of each machine is 0.05 Ω , calculate the efficiency of each machine for this particular load condition.	06	L3	02	1,2,9, 10,1
	c) List the different losses taking place in DC motor.	02	L1	02	1,2,9, 10,1

2. a) With necessary figure explain the operation of 4-point starter in DC motors. What is the advantage of 4-point starter over 3-point starter.
 b) A test on two coupled similar tramway motors, with their fields connected in series, gave the following test results when one machine acted as a motor and other as a generator.

Motor: Armature current 56 A; Armature voltage = 590 V

Generator: Armature current = 44 A; armature voltage = 400 V, field voltage drop = 40V

Resistance of each armature = 0.3Ω

- c) List the advantages of Hopkinson's test.
3. a) With a neat diagram, explain the working principle of the BLDC motor. What are its advantages and disadvantages over conventional DC motor.
 b) What is armature reaction? With suitable notations derive the expressions for demagnetizing and cross magnetizing MMF.
 c) A 2kW, 200 V DC shunt motor has an efficiency of 80%. Field current is 5 A. When operating at full load it produces demagnetizing MMF of 200 ampere turns. Calculate the number of additional turns required to neutralize the demagnetizing MMF.

08 L3 01 1,2,9, 10,1

06 L4 02 1,2,9, 10,1

02 L4 02 1,2,9, 10,1

08 L2 02 1,2,9, 10,1

06 L1 01 1,2,9, 10,1

02 L3 01 1,2,9, 10,1

Unit – II

4. a) Justify with proper reasoning the why MVA size alternators have stationary armature and rotating field system
 b) A 1200 kVA, 3300 volts, 50 Hz, three phase star connected alternator has an armature resistance of 0.25 per phase. A field current of 40 Amps produces a short circuit current of 200 A and an open circuit emf of 1100 volts line to line. Find the % regulation at full load 0.8 pf lagging by using EMF method. Also determine the power angle.
 c) Describe with relevant diagrams the process of synchronization of alternator to infinite bus using lamp bright method. What is meant by floating condition of a synchronous generator.
5. a) Explain with relevant diagrams, MMF method of pre determination of regulation of alternator.
 b) Analyze with relevant phasor diagrams, the various operation conditions of Synchronous generator connected to infinite bus at constant load with variable excitation. Plot V and Δ curve in generating mode
6. a) Derive EMF equation of an alternator
 b) Find the no load phase and line voltage of a 3-phase star connected 4-pole alternator from the following:
 Flux per pole = 0.12 Weber, Slots per pole per phase = 4
 Conductors/ slot = 4, Two-layer winding, with winding short pitched by two slots.

03 L2 03 1,2,9, 10,1

05 L3 03 1,2,9, 10,1

08 L2 04 1,2,9, 10,1

08 L2 03 1,2,9, 10,1

08 L4 04 1,2,9, 10,1

04 L2 03 1,2,9, 10,1

04 L3 03 1,2,9, 10,1

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- c) Enumerate the necessary conditions for parallel operation of synchronous generator
- d) Plot and explain power angle characteristics of a turbo alternator

04	L1	04	1,2,9, 10,1
04	L2	04	1,2,9, 10,1

Unit – III

7. a) Explain the experimental set up to conduct slip test to determine X_d & X_q of an alternator
- b) Derive the expression for power and draw & explain the power angle diagram of salient pole alternator. Comment on the value of load angle at peak power output.
8. a) Explain two different methods of starting synchronous motor.
- b) What is hunting? State its causes. How can it be minimized?

8	L2	05	1,2,9, 10,1
8	L3	05	1,2,9, 10,1
8	L2	05	1,2,9, 10,1
8	L2	05	1,2,9, 10,1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
